

Data Collection:

Lab 2:

Group No. 03

Instructor's Signature

Sudipto
19/2/26

Table 1:

Theoretical Value	Practical Value using DMM	% of Error
3.3 k Ω	3.22 k Ω	2.42 %
4.7 k Ω	4.57 k Ω	2.77 %
5.6 k Ω	5.45 k Ω	2.68 %
$R_T = R_1 + R_2 + R_3 = 13.6 \text{ k}\Omega$	13.24 k Ω	2.65 %

Table 2:

	Experimental readings	Theoretical values	% of Error
V_{in}	15.09 V	15 V	0.6 %
V_{R1}	3.66 V	3.64 V	0.55 %
V_{R2}	5.2 V	5.184 V	0.31 %
V_{R3}	6.20 V	6.176 V	0.39 %

Table 3: use data from table 2

Potential rise V_{in}	15.09 V	Are the voltage rises and drops equal?	
Sum of potential drops $V_{R1} + V_{R2} + V_{R3}$	3.66 + 5.2 + 6.20 = 15.06 V	☐ Equal	☒ Almost Equal ☐ Not Equal

Table 4:

	Experimental readings	Theoretical values	% of Error
V_{ab}	11.40 V	11.36 V	0.35 %

Questions

1. State the voltage division rule (VDR).
2. State the Kirchhoff's voltage law (KVL).
3. Showing all the steps, calculate R_{eq} . Compare with the experimental value.
4. Showing all steps, calculate the theoretical values in Table 2. Compare theoretical values to your experimental values and explain whether your circuit follows KVL or not.
5. Showing all the calculations, theoretically calculate V_{ab} . Compare with the experimental value and verify the voltage division rule at the terminal a-b.

MultiSim Simulation

- Simulate the KVL circuit you did in lab class. Fill the data tables. Verify KVL.